Toward a four-dimensional Yang–Mills construction

Notebook 08: a local flat minimizing extension

Research working document

16 October 2026

Abstract

We construct an explicit minimizing right inverse for the homogeneous flat Maxwell block average. A scalar gauge choice with a squared Laplacian denominator removes the remaining complex massless poles from the minimizing field itself. The resulting potential and every first fine difference have exponential locality in coarse-cell distance, uniformly in refinement. The potential has physical discrete $C^{1,\alpha}$ bounds and its curvature has C^{α} bounds for every fixed $\alpha < 1$. The scalar gauge matches the projection in the deposited homogeneous flat source formulas. The componentwise fine Laplacian is also uniformly bounded and local; this uses the physical averaging adjoint rather than an absolute second-derivative Fourier sum. The corrected bond average and its normalization are checked separately. This supplies local flat response legs, but does not prove a nonflat or independent-current analytic response theorem.

Status. Research proof checked by separate internal derivations and audits. No external certification or novelty claim is made. The explicit quadratic construction is proved below. The final source identification uses deposited transcriptions of the primary operator definitions and their stated minimizing property; the unavailable original CMP99b proof has not been independently reverified here.

1 The minimizing problem and the local norms

Let M be odd, $\eta = M^{-1}$, and use centered M-blocks. The normalized straight-string average and fine derivative are

$$(Q_M A)_\mu(n) = M^{-d-1} \sum_{s \in I_M^d} \sum_{j=0}^{M-1} A_\mu(n + (s + je_\mu)/M), \qquad I_M = \{-(M-1)/2, \dots, (M-1)/2\},$$

$$(D_{\eta,\nu} A)_\mu(x) = M[A_\mu(x + e_\nu/M) - A_\mu(x)], \qquad (d_\eta A)_{\mu\nu} = D_{\eta,\mu} A_\nu - D_{\eta,\nu} A_\mu.$$

Physical fine-volume norms have measure M^{-d}; unit coarse norms have counting measure. Define

$$q_M(B) = \inf_{Q_M A = B} \|d_\eta A\|_\eta^2 .$$

Notebook 07 derived its bounded holomorphic Fourier symbol and uniform exponential locality. We use its notation below and recall the relevant cancellation explicitly.

For fine values in a fixed physical cell, the local Hilbert norm is

$$\|v\|_{\mathrm{cell},2}^2 = M^{-d} \sum_{s \in I_M^d} |v(s/M)|^2.$$

The physical discrete C^α norm is the supremum norm plus

$$\sup_{s \neq t} \frac{|v(s/M) - v(t/M)|}{|(s-t)/M|^\alpha}.$$

The discrete $C^{1,\alpha}$ norm controls v, all full fine differences $D_{\eta,\nu}v$, and the C^α seminorms of these differences. Use a fixed physical collar whenever a fine difference or comparison crosses the cell boundary. Constants may depend on the collar size but never on M.

Theorem 1.1. *There is a linear extension $\mathcal{H}_M$ on the infinite lattice and every compatible periodic lattice such that*

$$Q_M\mathcal{H}_M = I, \qquad \|d_\eta \mathcal{H}_M B\|_\eta^2 = q_M(B).$$

It satisfies the scalar gauge condition defined in Section 5. Its reconstructed fine-cell symbol is holomorphic on a coarse momentum strip independent of M. For every fixed $0 < \alpha < 1$ there are constants $C_\alpha, c_\alpha > 0$, independent of M, such that its coarse-cell kernels obey

$$\begin{aligned}\|\mathcal{H}_M(n)\|_{\mathbb{C}^d \to C^{1,\alpha}_{\mathrm{cell}}} + \|(d_\eta \mathcal{H}_M)(n)\|_{\mathbb{C}^d \to C^{\alpha}_{\mathrm{cell}}} \\ + \|(\mathcal{L}_\eta \mathcal{H}_M)(n)\|_{\mathbb{C}^d \to L^{\infty}_{\mathrm{cell}}} \le C_\alpha e^{-c_\alpha |n|_1}.\end{aligned} \tag{1}$$

Here $\mathcal{L}_\eta = \sum_\nu D^*_{\eta,\nu} D_{\eta,\nu}$ acts componentwise on the potential; it is the scalar lattice Laplacian on each bond component, not the Maxwell curl precision. The same statement holds in physical cell Hilbert norms. Periodic kernels are exact periodizations and have uniform bounds in periodic coarse-cell distance.

Colors decouple at flatness; the scalar estimates act on each orthonormal color component. This theorem is about the stated full-block flat problem. Its physical source match is specified only in Section 8.

2 The alias quotient and its canceled symbol

For $r \in I_M^d$, put $p_r = z + 2\pi r$ and define

$$a_r = a_M(p_r), \quad a_M(p) = \prod_\nu \frac{1}{M} \sum_{s \in I_M} e^{ip_\nu s/M}, \quad \omega_{r,\mu} = a_M(p_r)\frac{1}{M}\sum_{j=0}^{M-1} e^{ip_{r,\mu} j/M}.$$

A superscript minus on a_r or ω_r means evaluation at $-p_r$, not variable-dependent complex conjugation. Set

$$u^\pm_{r,\mu} = M(e^{\pm i p_{r,\mu}/M} - 1), \quad P_r = \sum_\mu u^+_{r,\mu} u^-_{r,\mu} = 4M^2 \sum_\mu \sin^2(p_{r,\mu}/(2M)),$$

$$d^\pm_\mu = e^{\pm i z_\mu} - 1, \qquad b_\mu = d^+_\mu d^-_\mu.$$

Exact string telescoping gives

$$\omega_{r,\mu} u^+_{r,\mu} = a_r d^+_\mu. \tag{2}$$

The coarse average in a Fourier fiber is $B_\mu = \sum_r \omega_{r,\mu} A_{r,\mu}$. With physical-volume Parseval, fine fiber energies sum aliases without an extra M^d.

For real nonzero z, the fine Maxwell energy in an alias is

$$P_r |A_r|^2 - |(u^+_r)^* A_r|^2 = \min_\lambda P_r |A_r + u^+_r \lambda|^2.$$

Fine gradients average to coarse gradients by (2), and their scalar average is onto because $a_0 \neq 0$ on the central real zone. Thus the minimized form is the gauge quotient of

$$W = \operatorname{diag}\left(\sum_r \frac{\omega_{r,\mu}\omega^-_{r,\mu}}{P_r} \right),$$

namely

$$q_M = W^{-1} - W^{-1}d^+(d^-W^{-1}d^+)^{-1}d^-W^{-1}.$$

At $z = 0$, the constant fine field realizes any constant coarse field with zero energy, so $q_M(0) = 0$.

Write

$$A = a_0 a_0^-, \quad g_\mu = \omega_{0,\mu}\omega_{0,\mu}^-, \quad R_\mu = \sum_{r\neq 0} \frac{\omega_{r,\mu}\omega_{r,\mu}^-}{P_r}, \quad D_\mu = g_\mu + P_0 R_\mu, \quad h_\mu = D_\mu^{-1},$$

$$h = \mathrm{diag}(h_\mu), \qquad E = A^{-1} - \sum_\mu \frac{b_\mu R_\mu}{g_\mu D_\mu}.$$

The central instance of (2) gives $\sum_\mu b_\mu/g_\mu = P_0/A$, hence

$$d^- h d^+ = P_0 E, \qquad q_M = P_0 h - \frac{h d^+ d^- h}{E}. \tag{3}$$

This expression has no inverse power of the central massless denominator P_0.

3 The additional scalar denominator has a uniform domain

Define

$$R_{s,2} = \sum_{r\neq 0} \frac{a_r a_r^-}{P_r^2}, \qquad D_{s,2} = A + P_0^2 R_{s,2}. \tag{4}$$

The scalar weights are

$$t_{2,0} = \frac{a_0^-}{D_{s,2}}, \qquad t_{2,r} = \frac{P_0^2 a_r^-}{P_r^2 D_{s,2}} \quad (r \neq 0). \tag{5}$$

They satisfy $\sum_r a_r t_{2,r} = 1$. For real nonzero momentum, they minimize $\sum_r P_r^2|\lambda_r|^2$ at a prescribed scalar block average. Indeed the scalar covariance is $\sum_r a_r a_r^-/P_r^2 = D_{s,2}/P_0^2$ and the ordinary weighted least-squares formula gives (5).

Lemma 3.1. *There is a complex strip, with a positive real margin around the central Brillouin cube and width independent of M, on which $A, g_\mu, D_\mu, E, D_{s,2}, a_0$ are uniformly separated from zero. All these functions, their required inverses, $P_0, R_\mu, R_{s,2}$ and q_M are uniformly bounded and holomorphic there.*

Proof. Here are the uniform estimates behind the strip argument, so its use introduces no additional source assumption. Fix $\pi < K < 2\pi$ and a small imaginary width δ. For high aliases,

$$P_r(x) \geq c_K|r|^2, \qquad \Re P_r(x+iy) \geq P_r(x) - e^\delta \sum_\nu y_\nu^2, \quad |x_\nu| \leq K.$$

The first bound follows by taking distance from $2\pi M\mathbb{Z}$; for a nonzero retained coordinate that distance is at least $(2\pi - K)|r_\nu|$. Choose δ so the second term is smaller than half the high-alias gap. Discrete Parseval gives

$$\sum_r |a_r(x+iy)|^2 \leq e^{\sum|y_\nu|}, \quad \sum_r |\omega_{r,\mu}(x+iy)|^2 \leq e^{\sum|y_\nu| + 2|y_\mu|}.$$

Cauchy–Schwarz bounds the corresponding bilinear products with negative arguments. Thus both high remainders in (4) and the vector remainders are uniformly holomorphic and bounded, despite the growing number of aliases.

On the real central cube, the centered factor $\sin(t/2)/(M\sin(t/(2M)))$ is between $2/\pi$ and one. Consequently $A \geq (2/\pi)^{2d}$, $g_\mu, D_\mu \geq (2/\pi)^{2d+2}$, $D_\mu \leq 1 + d\pi^2/4$, and

$$E \geq \frac{4}{\pi^2(1 + d\pi^2/4)} \quad (z \neq 0), \qquad E(0) = 1.$$

Also $0 \leq R_{s,2} \leq 1/16$, so $D_{s,2} \geq A$. The enlarged real cube has the same estimates with weaker fixed positive constants: replace $2/\pi$ by $\sin(K/2)/(K/2)$ and the high gap by $4(2\pi - K)^2/\pi^2$.

Cauchy estimates on nested open tubes, retaining a real margin, give uniform derivatives. Integrating in imaginary directions first keeps $A, g_\mu, D_\mu, D_{s,2}$ nonzero. The formula for E is then uniformly bounded and holomorphic; a second shrinking and its real gap keep it nonzero as well. Since $A = a_0 a_0^-$ and each factor is bounded, both factors are bounded away from zero. All shrinkings are independent of M. No Cauchy disk leaves a previously controlled domain. For $M = 1$ the high sums are empty and the conclusion holds directly. □

4 The field and its exact central cancellation

For real nonzero momentum put

$$S = d^- hB, \qquad \theta = \frac{S}{P_0 E}, \qquad F_r = P_r^{-1} \operatorname{diag}(\omega_r^-) q_M B.$$

Define the proposed extension initially by

$$\mathcal{H}_r B = F_r + u_r^+ t_{2,r} \theta. \tag{6}$$

This formula explains the gauge choice. Its central term is defined on the complex null cone by the following cancellation:

$$(\mathcal{H}_0 B)_\mu = \omega_{0,\mu}^- h_\mu B_\mu + \frac{u_{0,\mu}^+}{a_0} \left(R_\mu h_\mu - \frac{P_0 R_{s,2}}{D_{s,2}} \right) \frac{S}{E}. \tag{7}$$

For a high alias,

$$\mathcal{H}_r B = P_r^{-1} \operatorname{diag}(\omega_r^-) q_M B + \frac{u_r^+ P_0 a_r^- S}{P_r^2 D_{s,2} E}, \qquad r \neq 0. \tag{8}$$

Proof of the cancellation. The cochain identity and $g_\mu h_\mu = 1 - P_0 R_\mu h_\mu$ give

$$\omega_{0,\mu}^- h_\mu d_\mu^+ = \frac{u_{0,\mu}^+}{a_0} (1 - P_0 R_\mu h_\mu).$$

Also $A/D_{s,2} = 1 - P_0^2 R_{s,2}/D_{s,2}$. Substitute these identities into the central part of (6). The two terms proportional to P_0^{-1} cancel, leaving exactly (7). For high aliases, cancel one P_0 between (5) and θ, obtaining (8). □

Lemma 3.1 now makes both expressions holomorphic. At zero, $\mathcal{H}_0(0) = I$ and every high alias vanishes, so constants are lifted to constants. No inverse of a zero harmonic eigenvalue has been introduced.

5 Constraint, minimal energy and scalar gauge

The Feynman minimizing field satisfies

$$\sum_r \operatorname{diag}(\omega_r) F_r = B - d^+\theta.$$

The added gradient averages to $d^+\theta$, by (2) and $\sum a_r t_{2,r} = 1$. Thus $Q_M \mathcal{H}_M = I$. Further, $d^- q_M = 0$, so

$$(u_r^+)^* F_r = (a_r^-/P_r) d^- q_M B = 0$$

on real nonzero fibers. The field F is fine transverse; its Maxwell and Feynman energies agree and equal $B^* q_M B$. The added gradient changes neither curl nor energy. This proves the minimizing identity in Theorem 1.1, including the separately specified constant zero mode.

Let Q_0 be the scalar block average and $\Delta = D_f^* D_f$ the positive fine scalar Laplacian. With the physical volume measures, $Q_0 Q_0^* = I$ and Q_0^* extends a coarse scalar as a blockwise constant fine scalar. For a positive coarse penalty a', define

$$G' = (\Delta + Q_0^* a' Q_0)^{-1}, \quad P = G' Q_0^* (Q_0 G'^2 Q_0^*)^{-1} Q_0 G', \quad R = I - P. \tag{9}$$

On a real finite regulator, P is the orthogonal projection onto $\operatorname{ran}(G' Q_0^*)$. In particular,

$$\operatorname{ran} R = \ker(Q_0 G') = (\Delta + Q_0^* a' Q_0) \ker Q_0 = \Delta \ker Q_0. \tag{10}$$

On a nonzero Bloch fiber, equivalently $\ker R = \operatorname{ran}(\Delta^{-1} Q_0^*)$. This follows by taking the orthogonal complement in (10); the Laplacian is invertible in such a fiber. At zero one uses (9), whose penalty treats scalar constants, rather than an undefined Δ^{-1}.

The divergence of the extension has the exact form

$$(D_f^* \mathcal{H}_M B)_r = P_r^{-1} a_r^- \frac{P_0 S}{D_{s,2} E} \qquad (z \neq 0). \tag{11}$$

The central expression is its canceled value $a_0^- S/(D_{s,2} E)$. Thus the divergence lies in $\operatorname{ran}(\Delta^{-1} Q_0^*)$, proving

$$R D_f^* \mathcal{H}_M = 0. \tag{12}$$

At zero the divergence is zero. The identities extend through the holomorphic formulas where appropriate; positivity and minimizing claims are made for real fields.

The gauge is unique among minimizing fields. In a nonzero fiber the difference of two energy minimizers is a fine gradient $u_r^+ \lambda_r$. Its zero average gives $\sum_r a_r \lambda_r = 0$. Therefore $\Delta\lambda \in \operatorname{ran} R$, while the gauge condition gives $R\Delta\lambda = 0$. Hence the difference vanishes. At zero, the coarse average fixes the fine constant harmonic one-form and all other zero-energy differences are high-alias gradients; the same argument removes them. Equivalently, (5) minimizes the divergence norm among the Maxwell energy minimizers. This is different from minimizing the potential norm itself.

6 Uniform fine derivative estimates

The sine bounds used in Lemma 3.1 give, for high aliases on a fixed smaller tube,

$$w(r) = \prod_\nu (1 + |r_\nu|)^{-1}, \qquad |a_r^-| \le C w(r), \qquad |\omega_{r,\mu}^-| \le \frac{C w(r)}{1 + |r_\mu|},$$

$$|u_r^+| \le C(1 + |r|), \qquad |P_r| \ge c(1 + |r|)^2.$$

These estimates are valid up to the edge of the fine Brillouin zone; they do not expand in r/M. Equation (8) therefore gives

$$\|\mathcal{H}_r\| \le \frac{Cw(r)}{(1+|r|)^2}, \qquad \left\|u^+_{r,\nu}\mathcal{H}_r\right\| + \left\|u^+_r \wedge \mathcal{H}_r\right\| \le \frac{Cw(r)}{1+|r|}. \tag{13}$$

For curl, the gradient term vanishes exactly and one may also use

$$u^+_r \wedge \mathcal{H}_r = P_r^{-1} u^+_r \wedge [\mathrm{diag}(\omega^-_r) q_M].$$

The central aliases and their first fine differences are bounded directly by (7).

On a dyadic shell $2^j \le |r|_\infty < 2^{j+1}$, choose a largest coordinate. Its harmonic sum over that shell is bounded, while each other coordinate contributes at most $C(j+1)$. Thus

$$\sum_{\text{shell } j} w(r) \le C_d (j+1)^{d-1}.$$

Combining with (13), for every $\alpha < 1$,

$$\sum_r (1+|r|)^\alpha \left[\|\mathcal{H}_r\| + \sum_\nu \left\|u^+_{r,\nu}\mathcal{H}_r\right\| + \left\|u^+_r \wedge \mathcal{H}_r\right\|\right] \le C_\alpha. \tag{14}$$

All constants are uniform in M. This absolute sum also proves uniform alias Hilbert operator bounds. It controls every full first fine derivative, unlike the simpler least-potential-norm gauge. This absolute-sum argument does not give the endpoint $\alpha = 1$ or pointwise second-derivative bounds. A separate physical-space identity does give the componentwise Laplacian bound below.

6.1 The componentwise Laplacian and the physical adjoint

Multiplication of the uncanceled formula by P_r gives

$$P_r \mathcal{H}_r B = \mathrm{diag}(\omega^-_r) q_M B + \mathcal{Z}_r B, \qquad \mathcal{Z}_r B = \frac{u^+_r a^-_r P_0 S}{P_r D_{s,2} E}. \tag{15}$$

The central expression is defined by cancellation as $\mathcal{Z}_0 B = u^+_0 a^-_0 S/(D_{s,2}E)$. For high aliases the established bounds give $\|\mathcal{Z}_r\| \le Cw(r)/(1+|r|)$. The same dyadic summation therefore proves uniform physical C^α bounds for the reconstructed remainder, on the fixed complex strip, for every $\alpha < 1$.

The first term of (15) is precisely $Q^*_M q_M B$ with the physical fine-volume adjoint. It should not be estimated by the absolute sum of its Fourier aliases. For $x = n + s/M$, $s \in I^d_M$, set $\beta_s = (s_\mu + (M-1)/2 + 1)/M$. Direct counting of the strings through that fine bond gives

$$(Q^*_M B)_\mu(n + s/M) = \beta_s B_\mu(n) + (1-\beta_s) B_\mu(n - e_\mu), \qquad 0 \le \beta_s \le 1. \tag{16}$$

The physical adjoint multiplies the discrete averaging coefficient by M^d. The two string multiplicities sum to M, so the two remaining weights sum to one. Transverse coarse labels are fixed by the cell containing the fine basepoint. This proves a uniform fine supremum bound and fixed coarse range for Q^*_M. Its reconstructed Bloch multiplier is $\beta_s + (1-\beta_s)e^{-iz_\mu}$, uniformly bounded on the strip. Consequently $Q^*_M q_M$ has uniform supremum-valued holomorphic cell symbols and exponential coarse-cell locality.

Equation (15) now proves the claimed L^∞ bound for $\mathcal{L}_\eta \mathcal{H}_M$ and its exponential locality by the same contour argument used below. The adjoint term may jump between neighboring transverse blocks, so the proof does not claim a uniform Hölder bound for the Laplacian or separate pointwise bounds for all second fine derivatives.

7 Fine-cell reconstruction and exponential locality

At a fine position $x = n + s/M$, reconstruct the symbol by

$$\mathcal{H}_{\mathrm{cell}}(z; s) = \sum_r e^{i(z+2\pi r)\cdot s/M} \mathcal{H}_r(z). \tag{17}$$

For real z, the matrix of exponential factors is unitary from alias counting-measure ℓ^2 to the cell norm with measure M^{-d}. For complex $z = x + iy$, multiplication by $e^{-y\cdot s/M}$ changes this norm and its inverse by at most $e^{\sum |y_\nu|/2}$. Thus no growing output-dimension factor appears in the physical Hilbert bound.

For two physical fine positions ξ, ξ' in a fixed cell or collar, the fixed strip gives

$$|e^{ip_r\cdot\xi} - e^{ip_r\cdot\xi'}| \le C_\alpha(1 + |r|)^\alpha |\xi - \xi'|^\alpha.$$

This follows from the derivative and bounded-exponential estimates and $\min(1, t) \le t^\alpha$. Apply (14) to (17), and to the symbols with factors $u^+_{r,\nu}$ or $u^+_r\wedge$. This proves bounded holomorphic cell symbols in physical discrete $C^{1,\alpha}$ for the potential and C^α for the curl. Collars cover differences across cell boundaries. A wrapped fine point carries the appropriate coarse Bloch factor, exactly as in the finite-difference symbol u^+_r.

The individual alias vector is not a fixed-index periodic symbol. On real momenta, a shift by 2π permutes aliases modulo M; the full scalar covariance, vector covariance, quotient and coarse derivative are periodic. Formula (6) therefore has that permutation law. In (17), a wrapped alias changes its phase by $e^{2\pi i s_\nu} = 1$. The reconstructed symbol is genuinely coarse-periodic. The canceled holomorphic formulas on enlarged momentum charts agree on real overlaps and hence glue by the identity theorem. This argument also applies to the fine derivative and curl symbols.

Finally shift each coarse Fourier contour within the common strip. Uniform operator bounds in the displayed cell norms give (1), including on fixed collars. The explicit kernel acts by

$$(\mathcal{H}_M B)(n + s/M) = \sum_m \mathcal{H}_M(n - m; s) B(m).$$

Exponential row sums give uniform coarse-ℓ^∞ to fine supremum and local Hölder bounds. The corresponding column sums use fine physical measure M^{-d}, which cancels the M^d fine positions per cell; weighted Schur gives the coarse-ℓ^2 to fine-volume-ℓ^2 bounds. Finite tori sample the same periodic symbols, so their kernels are exact periodizations. Absolute convergence gives uniform periodic-distance estimates, including small periods with oriented-cell multiplicities.

8 The homogeneous source identification

The scalar and bond operators must be checked separately. Raw I, lines 23909–23918, transcribes CMP99b (3.18)–(3.19): its scalar Q'_n is a uniform block average with adjoint transports. At flatness it is exactly Q_0. Raw I 23937–23944 transcribes CMP99b (3.24)–(3.25) as (9). The orthogonal projection calculation (10) therefore identifies the scalar gauge used above with the deposited homogeneous flat source gauge. This is not the simpler projection $I - Q_0^* Q_0$.

Raw I 24349–24373 transcribes the corrected one-step average of CMP98 (122)–(125). At the trivial configuration, all transports are identities, $Y = Y_x = 0$, and the four displayed correction brackets vanish. The remaining linear operator is exactly

$$(QA)(c) = \sum_{x\in B(c_-)} L^{-d-1} A([x, x']) = Q^{\mathrm{st}}_L A(c).$$

Raw I 47046–47048 gives the multilevel Q_j as compositions of these one-step operators. The centered digit and longitudinal-string bijections of Notebook 06 give their flat composition $Q^{\mathrm{st}}_{L^j}$. This is the corrected fine-to-coarse constraint, distinct from the rooted next-step $Q^\flat$ used to eliminate Gaussian pivot bonds.

For this homogeneous specialization, a source minimizing extension with $QH = I$ and $RD_f^*H = 0$ must equal $\mathcal{H}_M$ by the uniqueness argument above. If source notation instead reads $QH_0 = \ell^{-1}$, the normalized operator is $H_0\ell$, as recorded in Raw I 13638–13655. Omitting this normalization would compare different maps. The conclusion identifies the explicit deposited flat operator definitions and their stated minimizing property; it does not reproduce the original nonlinear minimizer theorem. The unavailable primary CMP99b text has not been independently verified in this check.

An equivalent algebraic correction helps compare gauges. If H is any minimizing right inverse with $QD_f = d_cQ_0$, then

$$H_{\mathrm{src}} = (I - D_fG'RD_f^*)H$$

is feasible, has the same curl energy, and satisfies the source gauge. Indeed $Q_0G'R = 0$ and $\Delta G'R = R$ follow from (9). Thus $QH_{\mathrm{src}} = I$ and $RD_f^*H_{\mathrm{src}} = 0$. Uniqueness identifies this correction with the explicit formulas proved local above; locality is supplied by those formulas, not assumed for a generic correction operator.

9 What the localized extension can support

The result supplies uniformly local fine potential, first-gradient and curl legs for the homogeneous flat source operator. For example, in a real differentiable family of self-adjoint stationary quadratic forms satisfying

$$QH = I, \qquad \Delta H = Q^*q, \qquad q = H^*\Delta H,$$

differentiation gives the exact identity

$$\dot q = H^*\dot\Delta H - H^*\dot Q^*q - q\dot QH.$$

It uses $Q\dot H = -\dot QH$ and requires no separate derivative bound on $\dot H$ in this first variation. Local variations of these stated coefficients can consequently be tested against the local flat legs just constructed.

This conditional identity by itself does not show that the deposited independent-current family has this stationary representation, or that its coefficient variations have the required weighted bounds. No arbitrary nonflat background, hole, frozen boundary or mixed history is covered here. No sharp cutoff is moved by an untracked gauge change. The full complex carrier domain remains a separate construction task. Notebook 09 now proves the stationary representation and a uniform complex current neighborhood for the homogeneous auxiliary family G^J_π, including all-order localized current responses. Notebook 11 separately controls the additional nonlinear constraint term and the full flat G^J_1 family in the physical-cell algebra along $M = L^j$. Notebook 10 uses the full inverse to prove locality of the projection onto feasible curvature variations and an all-order expansion for small curvature-metric perturbations at fixed averaging constraint.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b90867170b30fe10e`.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. 109 (1987), 249–301. Copy of the original article.